

MATH3171 (Mathematical Biology III)
YEAR 2025–26, MICHAELMAS TERM

ADDITIONAL PROBLEM SHEET 1

Topics covered: Chapters 1–2.

1. Practice for solving ODEs

(a) Solve

$$\frac{dy}{dx} = \frac{xy^3}{\sqrt{1+x^2}}, \quad (1)$$

subject to $y(0) = y_0$.

(b) Solve

$$x \frac{dy}{dx} = y(\log x - \log y), \quad (2)$$

subject to the initial condition $y(1) = A$.

(c) Solve

$$\frac{dy}{dx} = \frac{y^2}{x^2 + 3x + 2}. \quad (3)$$

2. 1D stability analysis

Consider the following 1D population models. In each case, find the equilibria and by drawing a phase plot, determine the stability of the equilibria:

(a) $\frac{dx}{dt} = x(x - 1),$

(b) $\frac{dx}{dt} = \sin(x),$

(c) $\frac{dx}{dt} = \frac{rx}{x + K},$ where $K, r > 0$.

Now, by letting $x = x_0 + \varepsilon x_1$, perform a formal linear stability analysis on the above models and confirm your earlier findings. You can either use the Taylor series result

$$\frac{dx}{dt} = f(x) \implies \frac{dx_1}{dt} = x_1 f'(x_0) \quad (4)$$

as seen in class, or you can derive the expansion yourself.

3. And fine-Allee... (Exam section B style)

Consider this version of the Allee equation describing growth of a bacterial population $x(t)$:

$$\frac{dx}{dt} = rx(K - x)(x - M), \quad (5)$$

where $r, K, M > 0$.

(a) Find the equilibria of this system.

- (b) Draw a phase plot for this model and determine the stability of each equilibrium, specifying how this depends on the relative size of K and M .
- (c) (i) Solve eq. (5) to show that the general solution $x(t)$ in the case $K = 1$ and $M = 2$ is

$$x(t) = 1 \pm \frac{\sqrt{Ae^{2rt}(Ae^{2rt} - 1)}}{1 - Ae^{2rt}} \quad (6)$$

for a constant A .

- (ii) By letting $t \rightarrow \infty$, confirm that the stable equilibria you determined before are indeed reached at infinite time.
- (iii) For which values of A is there a singularity in the solution $x(t)$ for $t > 0$?
- (d) Consider again the case $K = 1$, $M = 2$. Rewrite eq. (5) in the form

$$\frac{dx}{dt} = a_0x + a_1x^2 + a_2x^3, \quad (7)$$

for some a_0, a_1, a_2 . Now perform a formal linear stability analysis around each equilibrium x_0 by writing $x = x_0 + \varepsilon x_1$, and thus by finding the linearised solution $x_1(t)$, confirm the stability you found in part (b).

4. Time-averaged behaviour of Lotka–Volterra

Consider the normalised Lotka–Volterra equations,

$$\frac{dx}{dt} = x - xy, \quad (8a)$$

$$\frac{dy}{dt} = \gamma(-y + xy). \quad (8b)$$

Let T denote one period of the solution cycle, starting at some time t_0 .

- (a) Show that

$$\int_{t_0}^{t_0+T} \frac{1}{x} \frac{dx}{dt} dt = \int_{t_0}^{t_0+T} (1 - y) dt. \quad (9)$$

- (b) Using the solution periodicity, show that

$$T = \int_{t_0}^{t_0+T} y dt. \quad (10)$$

- (c) The time average, $\bar{f}$, of a function f over a period $t \in [t_0, t_0 + T]$ is given by

$$\bar{f} = \frac{1}{T} \int_{t_0}^{t_0+T} f dt. \quad (11)$$

Use this definition to show that

$$\bar{x} = 1 \quad \text{and} \quad \bar{y} = 1. \quad (12)$$

5. 2D stability analysis

Find the permissible equilibria of the following systems, and perform a linear stability analysis to determine whether they are asymptotically stable, unstable or degenerate.

- (a) This is at the easy end of the type of stability analysis you might encounter in an exam (but we have to start somewhere, right?):

$$\frac{dx}{dt} = -x - 2y, \quad (13a)$$

$$\frac{dy}{dt} = 4x - 5y. \quad (13b)$$

- (b) Try this for both $\pm$ cases:

$$\frac{dx}{dt} = xy \pm y^3, \quad (14a)$$

$$\frac{dy}{dt} = x \pm y^3. \quad (14b)$$

- (c) This is mathematically more challenging, and hence more like a standalone Section A exam question. But its biological relevance is perhaps less obvious:

$$\frac{dx}{dt} = -\sin(x^2 + y^2), \quad (15a)$$

$$\frac{dy}{dt} = x - y. \quad (15b)$$

6. Proof of periodicity of the solutions to Lotka–Volterra

We have shown in our notes that

$$\log y - y + \gamma(\log x - x) = C, \quad (16)$$

for some fixed constant C . The value of C is, of course, set by initial conditions (x_0, y_0) . We can therefore think about it as a function of all permissible initial condition pairs, $C = C(x_0, y_0)$, $x_0 > 0$, $y_0 > 0$.

- (a) Show that the function $C(x_0, y_0)$ has only one extremum, and find its coordinates. Further show that it is a (global) maximum.
- (b) [*Hard!*] Explain why this (and the differentiability of C on $x_0 > 0, y_0 > 0$) guarantees that the phase curves must be closed. (Hint: they are curves on a surface of the function C of fixed value.)

7. Variations on Lotka–Volterra (Exam 2021 & 2022)

The graphs on page 4 plot the populations and phase portraits of two species, $x(t)$ and $y(t)$, over time, t . Each graph in column **(G)** has the same initial condition and corresponds to one of the phase portraits in column **(P)**, and to one of the nondimensionalised models in column **(M)**. In the models, the constants $\beta, \gamma, \delta > 0$ wherever they appear, although they are not necessarily equal between graphs. The rows are numbered **(1)**–**(4)**.

Giving a good reason in each case, match the graphs to the phase portraits and the models. You should not need to find stability: you should be able to connect them by thinking about the models and their meaning alone.

For example, you could say that **(G1)** = **(P2)** = **(M1)**, if indeed this were correct.

Graphs, phase portraits and models for question 7

